KAIZEN MATHS

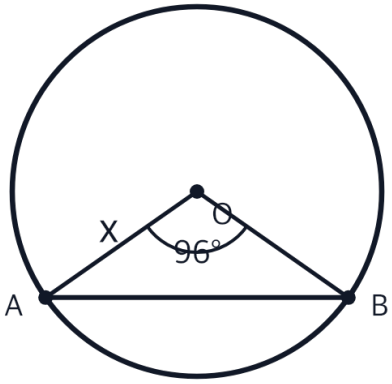
Lesson starter

Circle Theorems

Connected Angles

Name _____

SA1



OA and OB are radii. If ∠AOB = 96°, find ∠OAB.

GEO-475F-L2 · SA

Show your working

KAIZEN MATHS

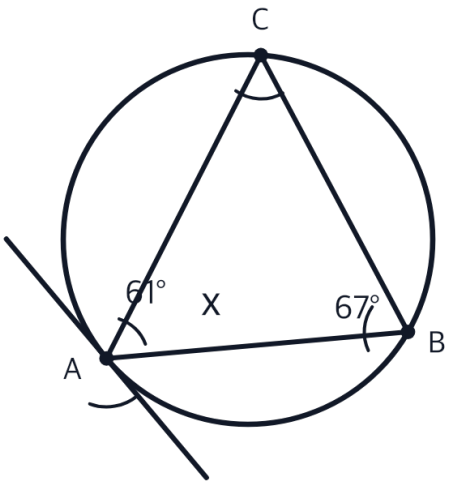
Lesson starter

Circle Theorems

Connected Angles

Name _____

SB1



The line at A is a tangent. The tangent-chord angle is 61° and ∠ABC = 67°. Find ∠CAB.

GEO-475F-L2 · SB

Show your working

KAIZEN MATHS

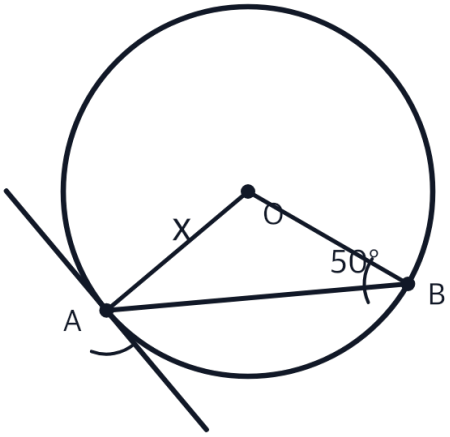
Lesson starter

Circle Theorems

Connected Angles

Name _____

SC1



The line at A is a tangent and OA = OB. If ∠ABO = 50°, find the angle between the tangent and chord AB.

GEO-475F-L2 · SC

Show your working

KAIZEN MATHS

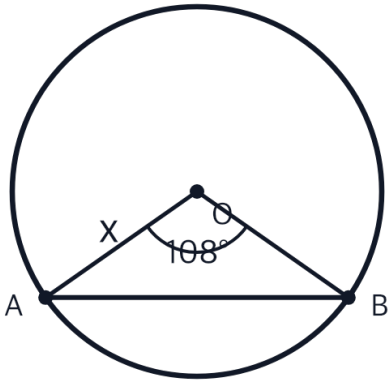
Lesson starter

Circle Theorems

Connected Angles

Name _____

SD1



OA and OB are radii. If ∠AOB = 108°, find ∠OAB.

GEO-475F-L2 · SD

Show your working

KAIZEN MATHS

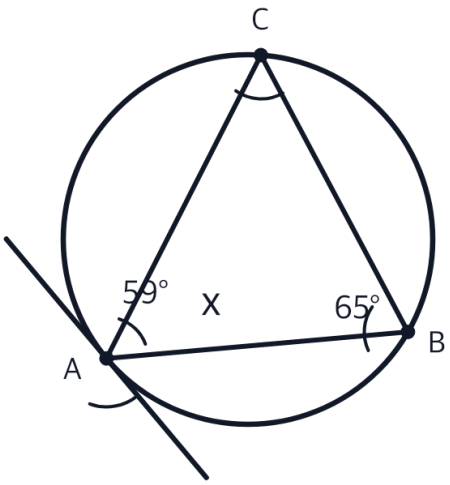
Exit ticket

Circle Theorems

Connected Angles

Name _____

EA1



The line at A is a tangent. The tangent-chord angle is 59° and ∠ABC = 65°. Find ∠CAB.

GEO-475F-L2 · EA

Show your working

KAIZEN MATHS

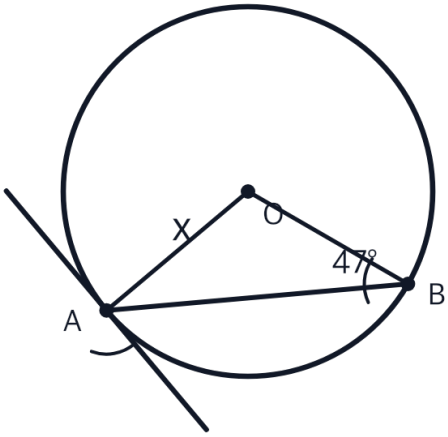
Exit ticket

Circle Theorems

Connected Angles

Name _____

EB1



The line at A is a tangent and OA = OB. If ∠ABO = 47°, find the angle between the tangent and chord AB.

GEO-475F-L2 · EB

Show your working

KAIZEN MATHS

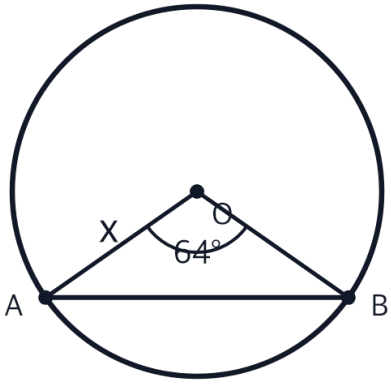
Exit ticket

Circle Theorems

Connected Angles

Name _____

EC1



OA and OB are radii. If ∠AOB = 64°, find ∠OAB.

GEO-475F-L2 · EC

Show your working

KAIZEN MATHS

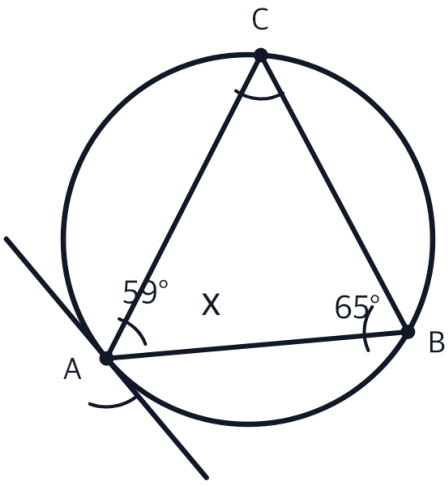
Exit ticket

Circle Theorems

Connected Angles

Name _____

EA1



The line at A is a tangent. The tangent-chord angle is 59° and ∠ABC = 65°. Find ∠CAB.

GEO-475F-L2 · EA

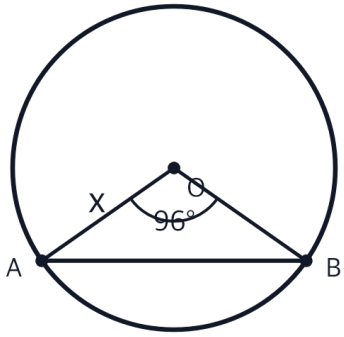
Show your working

Teacher answers & working

Circle Theorems · Connected Angles

GEO-475F-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



OA and OB are radii. If ∠AOB = 96°, find ∠OAB.

Answer:

42°

Working:

OA and OB are both radii, so triangle AOB is isosceles.

Therefore ∠OAB = ∠OBA.

Angles in triangle AOB add to 180°.

$\angle OAB + \angle OBA = 180° - 96°$

$2\angle OAB = 84°$

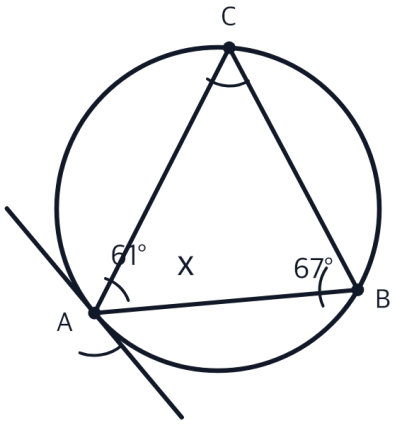
$\angle OAB = 84° \div 2 = 42°$

Teacher answers & working

Circle Theorems · Connected Angles

GEO-475F-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



The line at A is a tangent. The tangent-chord angle is 61° and ∠ABC = 67°. Find ∠CAB.

Answer:

52°

Working:

By the alternate segment theorem, the tangent-chord angle equals the angle in the alternate segment.

Therefore ∠ACB = 61°.

Angles in triangle ABC add to 180°.

$\angle CAB + \angle ABC + \angle ACB = 180°$

$\angle CAB + 67° + 61° = 180°$

$\angle CAB = 180° - 128°$

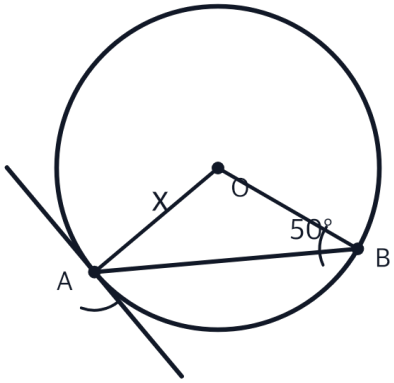
$\angle CAB = 52°$

Teacher answers & working

Circle Theorems · Connected Angles

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SC1



The line at A is a tangent and OA = OB. If ∠ABO = 50°, find the angle between the tangent and chord AB.

Answer:

40°

Working:

OA and OB are radii, so triangle AOB is isosceles.

Therefore ∠OAB = ∠ABO = 50°.

The tangent is perpendicular to radius OA, so the angle between the tangent and OA is 90°.

The angle between the tangent and chord AB is 90° - ∠OAB.

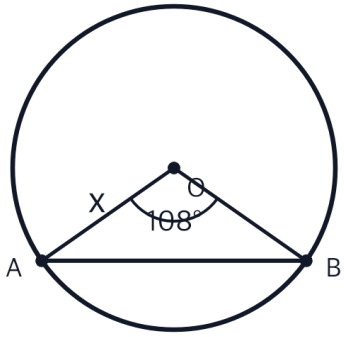
$x = 90° - 50° = 40°$

Teacher answers & working

Circle Theorems · Connected Angles

GEO-475F-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



OA and OB are radii. If ∠AOB = 108°, find ∠OAB.

Answer:

36°

Working:

OA and OB are both radii, so triangle AOB is isosceles.

Therefore ∠OAB = ∠OBA.

Angles in triangle AOB add to 180°.

$\angle OAB + \angle OBA = 180° - 108°$

$2\angle OAB = 72°$

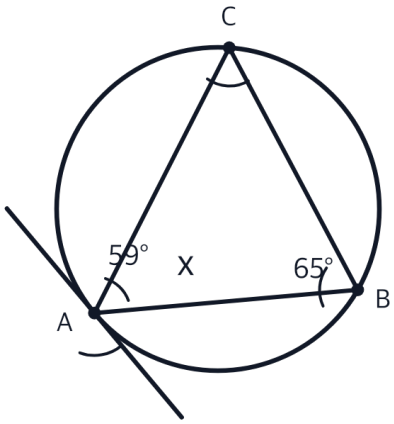
$\angle OAB = 72° \div 2 = 36°$

Teacher answers & working

Circle Theorems · Connected Angles

GEO-475F-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



The line at A is a tangent. The tangent-chord angle is 59° and ∠ABC = 65°. Find ∠CAB.

Answer:

56°

Working:

By the alternate segment theorem, the tangent-chord angle equals the angle in the alternate segment.

Therefore ∠ACB = 59°.

Angles in triangle ABC add to 180°.

$\angle CAB + \angle ABC + \angle ACB = 180°$

$\angle CAB + 65° + 59° = 180°$

$\angle CAB = 180° - 124°$

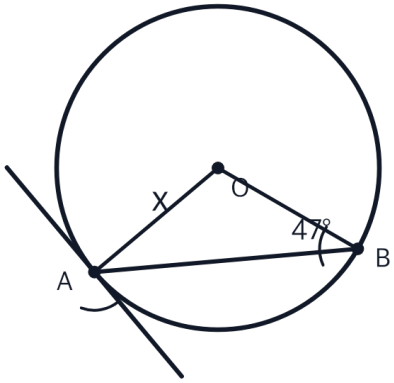
$\angle CAB = 56°$

Teacher answers & working

Circle Theorems · Connected Angles

GEO-475F-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



The line at A is a tangent and OA = OB. If ∠ABO = 47°, find the angle between the tangent and chord AB.

Answer:

43°

Working:

OA and OB are radii, so triangle AOB is isosceles.

Therefore ∠OAB = ∠ABO = 47°.

The tangent is perpendicular to radius OA, so the angle between the tangent and OA is 90°.

The angle between the tangent and chord AB is 90° - ∠OAB.

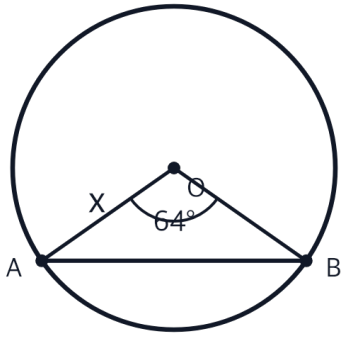
$x = 90° - 47° = 43°$

Teacher answers & working

Circle Theorems · Connected Angles

GEO-475F-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



OA and OB are radii. If ∠AOB = 64°, find ∠OAB.

Answer:

58°

Working:

OA and OB are both radii, so triangle AOB is isosceles.

Therefore ∠OAB = ∠OBA.

Angles in triangle AOB add to 180°.

$\angle OAB + \angle OBA = 180° - 64°$

$2\angle OAB = 116°$

$\angle OAB = 116° \div 2 = 58°$